

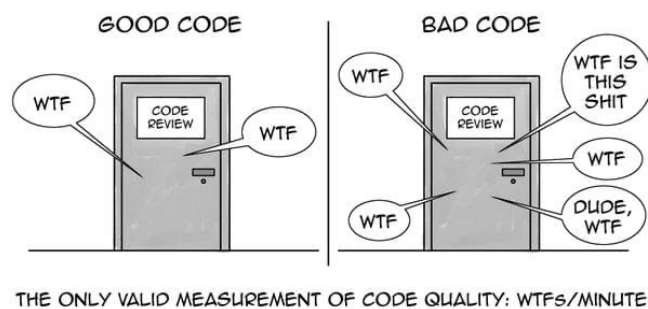
Topics Covered

- The topics in this session will focus on:
 - Version control to help track and manage changes to your code and data.
 - Coding practices for staying organized in large projects.



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Source code



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Learning Objectives

- The learning objectives of this session are:
- * Understand the basics of version control and can use `git` to track changes to your code.
- * Knowledge of how to package Python code into a library and how to organize your code for reuse.
- * Understand different coding practices and how to use them to improve the quality of your code.
- * Can use `dvc` to version control data.



Why Version Control?

- Version control is essential for working on code collaboratively.
 - Keep track of changes in the code.
 - Know who made changes and when.
 - Understand what specific changes were made.
 - For more information, see this page: <https://git-scm.com/book/en/v2/Getting-Started-What-is-Git>



GitHub vs Git

- GitHub is a platform for hosting Git repositories.
- Other platforms include Bitbucket and GitLab.
- This course requires using Git and GitHub.
- Create a public repository for your code and use Git for version control.



Initial Git Setup

1. [Install Git](#) on your computer.
2. Create a GitHub account.
3. Set up credentials on your local machine:
 - `git config credential.helper store`
 - `git config --global user.email <email>`
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What Does Git Stand For?

- The name "git" was given by Linus Torvalds when he wrote the first version.
- Described as:
 - * Random three-letter combination.
 - * Stupid, contemptible, despicable.
 - * Global information tracker.
 - * Goddamn idiotic truckload of sh*t.
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Git Overview

- Version control is a system of nodes connected by lines.
- Each node, called a *commit*, is identified by a unique hash.
- Commits store the state of the code at specific times.



Commit Workflow

- `git add` moves changes to the *staging area*.
- `git commit` creates a commit locally.
- `git push` uploads the commit to the remote repository.

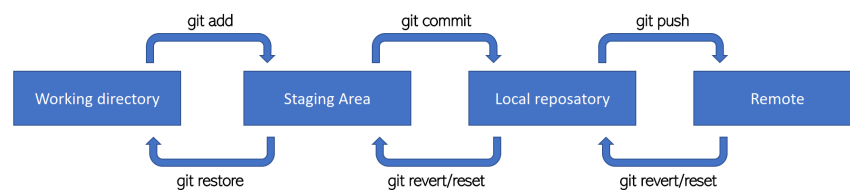


Figure: Basic Git Workflow



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Branching in Git

- Branches allow multiple versions of code to exist simultaneously.
- Useful for teams working on different features or bug fixes.

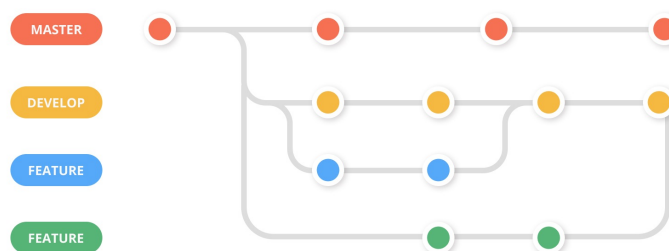


Figure: Git Branches



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How do you know if a directory is a git repository?

- Check if there is a ".git" directory.
- Alternatively, use the command:
 - `git status`
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What is gitignore used for?

- Instructs git on which files to ignore.
- Useful for files not part of the codebase (e.g. data files) or files containing sensitive information (e.g. .env files).



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Syncing devel with main

- Commands to sync *devel* with *main*:

```
git checkout main  
git pull  
git checkout devel  
git merge main
```

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Best practices for version control

- Use descriptive commit messages.
- Make each commit a logical unit.
- Incorporate others' changes frequently.
- Share your changes regularly.
- Coordinate with co-workers.
- Avoid committing generated files.



Additional thing

- GitHub's built-in editor can be enabled by changing the URL:
`https://github.com/username/repository`
- to
`https://github.dev/username/repository`

